FWR: A Minimal Observational Language for Structural Survival Dynamics

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Abstract

We present FWR (Flow–Structure–Resonance), a minimal discrete-time dynamical system intended not as a predictive scientific theory but as a *minimum observational language*: a small, fixed set of questions (inflow, structure, alignment, efficiency, history) that can be asked of any bounded system undergoing episodes of stability, crisis, and structural reorganization. We give a fully closed axiomatic specification of the system's internal state transitions, prove that the accumulated-history state variable is uniformly bounded regardless of the alignment sign or crisis frequency, and specify an endogenous crisis-detection rule that triggers discrete structural pivots. We are explicit about what the model does *not* determine: the post-pivot structure value is treated as an exogenous boundary condition, and we show why a naive endogenous optimization of this value produces a self-defeating collapse mode. We close with a discussion of the model's intended scope and open empirical questions.

1 Introduction

Most formal models of system health conflate two distinct claims: a claim about *what quantities matter* for a system's persistence, and a claim about *how those quantities evolve in physical time*. FWR is offered as an answer to the first question only. It does not claim to predict the trajectory of any particular real system (a company, a person's skill development, a training run); it claims that five quantities — inflow F, structural capacity W, alignment C, efficiency η, and accumulated history T — are sufficient to define a coherent, internally consistent notion of *survival* for a bounded discrete-time system, and that asking about these five quantities in sequence ($F \to W \to C \to \eta \to T$) is a useful diagnostic regardless of whether the underlying system obeys FWR's specific functional forms.

The axiomatic content of this paper — the specific functions $\Phi = \min(F, W)$, $g(T) = 1 + T/(1 + \alpha T)$, and the transition rule governing T and W — should be read as *one* minimal instantiation of this observational language, chosen for its closed-form tractability and its provable boundedness properties, not as an empirical claim about any specific domain.

1.1 Relation to prior versions

This specification is the result of an iterative adversarial-review process in which each successive version was checked for: dimensional consistency, undefined free parameters, ambiguity between alternative functional forms, sign consistency between mathematical structure and its stated interpretation, and completeness of the state-transition function at discontinuities (structural pivots). We summarize the resolved issues in Section 10 for transparency, since several of them reflect non-obvious failure modes that a reader extending this model is likely to re-encounter.

2 State Space and Variables

Definition 1 (Variable classification)**.** *The system distinguishes three kinds of quantities:*

(i) ***External inputs****, observed at every discrete step* $n \in \mathbb{N}$ *and not governed by any transition rule internal to the model:* $F(n) \in [0, 1]$ *(inflow) and* $C(n) \in [-1, 1]$ *(alignment,* $C = \cos\theta$ *for some angle* θ *between the system's actual and required orientation).*

(ii) ***Internal state variables****, governed by a transition function defined below:* $T(n) \in [0, \infty)$ *(history) and* $W(n) \in [0, 1]$ *(structural capacity).*

(iii) ***Parameters****, fixed for a given system instance:* $\alpha > 0$ *(history saturation coefficient, an internal structural constant), and* $\eta \in (0, 1]$*,* $\gamma \in (0, 1]$*,* $\beta \in [0, 1]$ *(efficiency, forgetting rate, and legacy-compatibility coefficient, treated as externally set control parameters).*

Remark 1. *An earlier version of this specification treated* W *as a purely external input on the grounds that its value is, most of the time, determined outside the system. This is inconsistent: if* W *is reset at a pivot and then* persists *until the next pivot, the system must be remembering that reset value, which is precisely the definition of state.* W *is therefore classified as an internal state variable with a* piecewise-constant *transition rule (Definition 5), not as an external input.*

3 Output and Crisis Detection

Definition 2 (Potential and history amplification)**.**

$$P(n) = \Phi(F(n), W(n)) = \min(F(n), W(n)), \qquad g(T) = 1 + \frac{T}{1 + \alpha T}.$$

Proposition 1 (Saturation of g)**.** *For* $\alpha > 0$*,* g *is strictly increasing on* $[0, \infty)$ *and*

$$\lim_{T \to \infty} g(T) = 1 + \frac{1}{\alpha}, \qquad g'(T) = \frac{1}{(1 + \alpha T)^2} > 0.$$

Proof. Direct computation; verified symbolically. g is a Möbius-type function of T composed with an affine shift, hence monotone with a finite limit as $T \to \infty$. □

Definition 3 (Existence output)**.**

$$E(n) = P(n) \cdot C(n) \cdot \eta \cdot g(T(n)) = \min(F(n), W(n)) \cdot C(n) \cdot \eta \cdot \left[1 + \frac{T(n)}{1 + \alpha T(n)}\right].$$

$E(n)$ is signed: $E(n) > 0$ when the system is aligned ($C(n) > 0$) and producing output; $E(n) < 0$ when the system is misaligned ($C(n) < 0$), which we interpret as a self-destructive or counter-resonant regime.

Definition 4 (Crisis condition)**.**

$$\text{crisis}(n) \;:\equiv\; E(n) \leq \gamma \cdot T(n).$$

This says: the system is in crisis at step n exactly when its current output fails to outpace the decay of its accumulated history. Because $E(n)$ can be negative while $\gamma T(n) \geq 0$, any misaligned state ($C(n) < 0$) is automatically a crisis state, independent of magnitude.

4 Transition Function

Definition 5 (Unified transition rule)**.** *For each n, exactly one of the following two branches applies:*

$$\textit{if } \text{crisis}(n): \quad \begin{cases} T(n+1) = \beta \cdot T(n) \\ W(n+1) = W_k^* \end{cases} \qquad \textit{if } \neg\,\text{crisis}(n): \quad \begin{cases} T(n+1) = T(n) + |E(n)| - \gamma \cdot T(n) \\ W(n+1) = W(n) \end{cases}$$

where $W_k^* \in [0,1]$ is the structural value assigned at the k-th crisis step (see Section 7).

Remark 2. *Earlier drafts of this model introduced left/right-limit notation (p_k^-, p_k^+) to describe a pivot as an instantaneous event nested inside a discrete step $n = p_k$, borrowed from the Filippov framework for continuous-time discontinuities. In discrete time this notation is unnecessary and creates a genuine ambiguity: it leaves undefined whether $E(n)$ should be evaluated pre- or post-pivot, and whether the following step's transition composes both the pivot rule and the ordinary rule or replaces one with the other. Definition 5 removes this ambiguity by construction: each step applies exactly one branch, evaluated using the state at the start of that step.*

A continuous-time analogue for T alone (holding W, F, C fixed between crises) is

$$T(\tau) = \int_0^\tau |E(\tau')|\, e^{-\gamma(\tau-\tau')}\, d\tau',$$

where τ is a dimensionless time variable; we do not develop the continuous-time pivot structure in this paper.

5 Boundedness

Theorem 1 (Uniform boundedness of history)**.** *For all n,*

$$T(n+1) \leq (1-\gamma)\, T(n) + \left(1 + \frac{1}{\alpha}\right),$$

and consequently

$$\limsup_{n\to\infty} T(n) \;\leq\; \frac{1 + \frac{1}{\alpha}}{\gamma}.$$

Proof. By Proposition 1, $g(T(n)) \leq 1 + 1/\alpha$ for all $T(n) \geq 0$, uniformly in n — this bound does not depend on $T(n)$ itself, so no circularity arises. Since $F(n), \eta, |C(n)| \leq 1$,

$$|E(n)| = \min(F(n), W(n)) \cdot |C(n)| \cdot \eta \cdot g(T(n)) \;\leq\; 1 \cdot 1 \cdot 1 \cdot \left(1 + \frac{1}{\alpha}\right) =: E_{\max}.$$

On non-crisis steps, $T(n+1) = T(n) + |E(n)| - \gamma T(n) \leq (1-\gamma)T(n) + E_{\max}$. On crisis steps, $T(n+1) = \beta T(n) \leq T(n) \leq (1-\gamma)T(n) + E_{\max}$ whenever $\beta \leq 1 - \gamma + E_{\max}/T(n)$, which holds trivially since $\beta \leq 1$ and the right-hand side exceeds 1 for any $T(n) < E_{\max}/\gamma$; for $T(n) \geq E_{\max}/\gamma$ the crisis branch strictly decreases T, so the bound is preserved in either branch. Iterating the linear recursion with $\gamma \in (0,1]$ gives the stated lim sup. □

Remark 3. *The requirement $\gamma > 0$ (rather than $\gamma \in [0,1]$) is essential: at $\gamma = 0$ the decay term vanishes and $T(n)$ can grow without bound even though $|E(n)|$ is uniformly bounded, since a bounded positive increment added every step without decay still diverges.*

6 Pivot Sequence and Stabilization

Definition 6 (Pivot sequence). *The pivot sequence is the (possibly finite, possibly empty) increasing sequence* $p_1 < p_2 < \cdots$ *enumerating* $\{n : \text{crisis}(n)\}$. *This is a derived quantity, not an independent axiom: it follows directly from Definition 5 once a trajectory is fixed.*

Definition 7 (Stabilization). *The system is said to* stabilize *if*

$$\exists K \geq 1 \text{ such that } \forall n \geq K,\ \neg\,\text{crisis}(n),$$

equivalently $E(n) > \gamma T(n)$ *for all* $n \geq K$. *In this case the pivot sequence is finite and the system persists indefinitely under the non-crisis branch alone.*

Theorem 1 guarantees T does not diverge whether or not the system stabilizes; it does not by itself guarantee stabilization occurs, since $E(n)$ depends on the exogenous sequences $F(n), C(n)$, which are unconstrained.

7 Scope: What FWR Does Not Determine

Definition 5 leaves W_k^* — the structural value assigned at each crisis step — as an exogenous input. We take this to be a deliberate scope boundary rather than an incompleteness to be patched.

Proposition 2 (Naive endogenous optimization is self-defeating). *Suppose* W_k^* *is chosen to maximize the post-pivot output,*

$$W_k^* = \underset{W \in [0,1]}{\arg\max}\ C(p_k) \cdot \min(F(p_k), W)$$

(the η *and* $g(T)$ *factors are independent of* W *and do not affect the arg max). Then:*

- *if* $C(p_k) > 0$, *any* $W \geq F(p_k)$ *is optimal (a degenerate, non-unique corner solution);*
- *if* $C(p_k) < 0$, *the unique optimum is* $W_k^* = 0$.

In the latter case, $P(n) = \min(F(n), 0) = 0$ *for all subsequent* n, *forcing* $E(n) = 0$ *for all subsequent* n. *Since* $E(n) = 0 \leq \gamma T(n)$ *whenever* $T(n) > 0$, *the system remains in* crisis *at every subsequent step, re-applying this same rule and never escaping.*

This shows that a locally reasonable-looking optimization objective produces a global collapse trap precisely in the regime (misalignment, $C < 0$) where a recovery mechanism is most needed. We do not consider this a defect to be patched within the model; we take it as evidence that choosing W_k^* requires information — about *why* the system is misaligned, not merely *that* it is — which lies outside the five quantities (F, W, C, η, T) that FWR tracks. Accordingly:

Axiom 1 (Exogeneity of structural choice). $W_k^* \in [0, 1]$ *is supplied by a process external to the model (a designer, a coach, a higher-level controller, or a separate optimization procedure not specified here). FWR evaluates the consequences of a given choice of* W_k^* *for the system's subsequent trajectory; it does not prescribe* W_k^*.

Remark 4 (Terminology). *For this reason we do not describe FWR as a* closed *dynamical system in the sense of having no external inputs —* F, C, *and* W_k^* *all enter from outside. We instead describe it as a* deterministic *dynamical system: given any realization of the external inputs, the state trajectory* $(T(n), W(n))_{n \geq 0}$ *is uniquely determined by Definition 5. Determinism concerns the completeness of the transition rule; closure would concern the absence of external inputs, which this model does not claim.*

8 Full Specification

$$\text{State: } (T(n), W(n)) \in [0, \infty) \times [0, 1]. \tag{1}$$

$$E(n) = \min(F(n), W(n)) \cdot C(n) \cdot \eta \cdot \left[1 + \frac{T(n)}{1 + \alpha T(n)}\right]. \tag{2}$$

$$\text{crisis}(n) :\equiv E(n) \leq \gamma T(n). \tag{3}$$

$$\text{If crisis}(n): \; T(n+1) = \beta T(n), \; W(n+1) = W_k^*.$$

$$\text{Else: } T(n+1) = T(n) + |E(n)| - \gamma T(n), \; W(n+1) = W(n). \tag{4}$$

$$\limsup_{n\to\infty} T(n) \leq \left(1 + \tfrac{1}{\alpha}\right)/\gamma. \tag{5}$$

9 Interpretation Registry

Condition	Interpretation
$C(n) > 0,\ E(n) > \gamma T(n)$	Aligned, persisting
$C(n) > 0,\ E(n) \leq \gamma T(n)$	Aligned but output insufficient; crisis
$C(n) = 0$	Neutral / inert output
$C(n) < 0$	Counter-resonant; always a crisis state
$\exists K,\ \forall n \geq K,\ \neg\text{crisis}(n)$	Stabilized

10 Summary of Resolved Issues

The following issues were identified and resolved across the model's revision history; we record them because each reflects a general class of error likely to recur in similar minimal-language constructions.

1. **Dimensional consistency.** An earlier form of $g(T)$ added a dimensionless constant to a term with units of T. Resolved by declaring all state variables and time steps dimensionless from the outset, rather than after the fact.

2. **Dangling parameters.** An earlier definition $\eta = 1/(1+\mu)$ introduced μ without defining it. Resolved by treating η as a directly specified control parameter.

3. **Underdetermined functional form.** $\Phi(F, W)$ was left as a choice between $\min(F, W)$ and $\sqrt{FW}$. Fixed to min for its bottleneck interpretation; the geometric-mean alternative is not excluded for other domains but is not part of the core specification.

4. **Sign of E.** Whether a negative existence output should be interpreted as a modeling artifact or a substantive state (self-destruction) was left implicit. Resolved via the interpretation registry above.

5. **Adaptive forgetting rate.** An earlier version allowed $\gamma(t) \to 0$, which breaks the boundedness argument in Theorem 1. Resolved by fixing $\gamma \in (0, 1]$ as a constant in the core specification.

6. **State/input classification.** W was initially classified as a pure external input, which is inconsistent with its persistence across steps between pivots (Section 7). Resolved by reclassifying W as an internal state variable with a piecewise-constant transition rule.

7. **Pivot-trigger existence and multiplicity.** A single pivot time $p = \min\{n : \text{crisis}(n)\}$ does not account for repeated structural transitions, nor for the possibility that the defining set is empty (i.e., the system stabilizes and no further pivot occurs). Resolved by defining the pivot sequence as a derived enumeration of $\{n : \text{crisis}(n)\}$, which may be finite, rather than as an independently asserted single value.

8. **Same-step discontinuity handling.** Left/right-limit notation for pivots (p_k^-, p_k^+) is well suited to continuous time but ambiguous once the state is indexed by an integer n; in particular, it leaves unspecified whether the pivot rule and the ordinary update rule compose within one step or replace one another. Resolved by the single-branch transition rule of Definition 5.

9. **Naive endogenization of W_k^*.** Treating W_k^* as the output of a local optimization is not scope-neutral: as shown in Proposition 2, the natural objective produces a permanent collapse when $C(p_k) < 0$. Resolved by explicitly scoping W_k^* as an exogenous boundary condition rather than attempting a general-purpose endogenous rule.

11 Discussion and Open Questions

FWR's claim to usefulness rests on the observational sequence $F \to W \to C \to \eta \to T$ being a productive set of questions to ask of a system in difficulty, independent of whether that system's true dynamics match Definition 5 exactly. This claim is not addressed by the results in this paper, which are purely internal to the axiomatic specification. Two questions of that kind seem to us the most pressing for further work:

- **Measurement.** $F(n)$, $W(n)$, and $C(n)$ are treated here as directly observed. In any concrete application (organizational diagnosis, skill-development coaching, or diagnosing failure modes in a learning system) these quantities must be estimated, and the sensitivity of $\text{crisis}(n)$ to estimation error in $C(n)$ in particular — since it alone can flip the sign of $E(n)$ — has not been characterized.

- **Threshold sensitivity.** The crisis condition $E(n) \leq \gamma T(n)$ uses the same γ that governs decay. It is not yet established how often this condition would trigger under realistic noisy trajectories of F, C, and whether the resulting pivot frequency is a useful diagnostic signal or merely an artifact of parameter choice.

We leave both to empirical follow-up work outside the scope of this specification.